

Movement of a metal ball in a wire field — Solution

X23 (2023) — English translation

A1

0.80 pts

A metal ball of radius R is placed in a uniform electric field $\vec{E}_0$. Find the dipole moment $\vec{p}$ acquired by the ball. Express your answer in terms of $\vec{E}_0$, R and the electric constant ϵ_0 .

The electric field inside the ball is zero, and outside the ball it is the sum of the external field and the dipole field:

$$\vec{E} = \vec{E}_0 + \frac{1}{4\pi\epsilon_0} \left(\frac{3(\vec{p} \cdot \vec{r})\vec{r}}{r^5} - \frac{\vec{p}}{r^3} \right)$$

On the surface of the ball, the electric field is directed perpendicular to the surface. From here:

Answer: \textbf{Answer:}

$$\vec{p} = 4\pi\epsilon_0 R^3 \vec{E}_0$$

A2

0.50 pts

Let the ball be located in an electric field directed along the z axis, the intensity of which varies according to the law:

$$\vec{E} \approx (E_0 + kz) \vec{e}_z$$

where E_0 and k — известные величины, moreover $kR \ll E_0$. Find силу F_z , действующую на ball/sphere. Answer выtimesandme via/through E_0 , k , R and электрическую постоянную ϵ_0 .

By definition, a dipole is two point charges $+q$ and $-q$ located at a distance l from each other. Let $\vec{r}$ be the radius vector of the negative charge, and $\vec{l}$ be the radius vector of the positive charge relative to the negative one. Then for the interaction force we have:

$$\vec{F} = q \left(\vec{E}(\vec{r} + \vec{l}) - \vec{E}(\vec{r}) \right)$$

If electric field направлено вдоль дипольного момента системы, то, since диполь ориентирован вдоль оси z , force, действующая nnotro, направлена вдоль оси z . If диполь можно считать точечным

$$E_z(\vec{r} + \vec{l}) - E_z(\vec{r}) = \frac{dE_z}{dz} \cdot l$$

Hence for силы we get:

$$F_z = p_z \frac{dE_z}{dz}$$

В нашей задаче since $kR \ll E_0$ - модель электрического диполя хорошо описывает electric field ball/spherea. Substituting expression for p_z and $\frac{dE_z}{dz}$, we find:

Answer: \textbf{Answer:}

$$F_z = 4\pi R^3 \epsilon_0 E_0 k$$

A3**0.20 pts**

Find the magnitude of the electric field of the wire E_n at a distance r from it. Express your answer in terms of λ , r and the electric constant ϵ_0 .

From Gauss's theorem we have:

$$2\pi r L E_n = \frac{q}{\epsilon_0} = \frac{\lambda L}{\epsilon_0}$$

whence:

Answer: \textbf{Answer:}

$$E_n = \frac{\lambda}{2\pi\epsilon_0 r}$$

A4**0.50 pts**

Find the magnitude and direction (attraction or repulsion) of the force F acting on the ball from the side of the wire at a distance r from it. Express your answer in terms of λ , R , r and the electric constant ϵ_0 .

At $r \gg R$ the values E and $\frac{dE}{dr}$ of the wire fields on the ball size scale can be considered constant. In this approximation, the dipole moment of the ball is equal to:

$$p = \frac{2\lambda R^3}{r}$$

Force, действующая на ball/sphere, equals:

$$F_r = p \frac{dE_r}{dx}$$

whence:

Answer: \textbf{Answer:} The force of attraction is:

$$F = \frac{\lambda^2 R^3}{\pi\epsilon_0 r^3}$$

A5**0.40 pts**

Show that the mechanical potential energy W_p of a ball in the field of a wire at a distance r between the wire and the center of the ball is equal to:

$$W_p = -\frac{A}{r^2}$$

Find A . Answer выtimesandme via/through λ , R and электрическую постоянную ϵ_0 .

For potential energy we have:

$$W_p = -\int_r^\infty F dr$$

Integrating, we find:

$$W_p = -\frac{\lambda^2 R^3}{2\pi\epsilon_0 r^2}$$

Hence parameter A is equal to:

Answer: \textbf{Answer:}

$$A = \frac{\lambda^2 R^3}{2\pi\epsilon_0}$$

B1

0.60 pts

Find the minimum distance $r_{\min}$ between the center of the ball and the wire during motion, assuming that no collision occurs. Express your answer in terms of A , b , m and v_0 .

From the law of conservation of energy we have:

$$\frac{mv^2}{2} - \frac{A}{r^2} = \frac{mv_0^2}{2}$$

Let us represent квадрат скорости в следующем виде:

$$v^2 = \dot{r}^2 + r^2\dot{\varphi}^2$$

From the law сохранения angular momentum we get:

$$L = mr^2\dot{\varphi} \Rightarrow \dot{\varphi} = \frac{v_0 b}{r^2}$$

Комбинируя с законом сохранения энергии, we get:

$$\frac{m\dot{r}^2}{2} + \frac{mv_0^2 b^2}{2r^2} - \frac{A}{r^2} = \frac{mv_0^2}{2}$$

Минимальное distance до проwater достигается for/at $\dot{r} = 0$. From here we get:

Answer: \textbf{Answer}:

$$r_{\min} = \sqrt{b^2 - \frac{2A}{mv_0^2}}$$

B2

0.20 pts

Find v_{crit} . Express your answer in terms of A , b and m .

The radical expression must be non-negative. From here we get:

Answer: \textbf{Answer}:

$$v_{\text{crit}} = \frac{1}{b} \sqrt{\frac{2A}{m}}$$

C1

0.40 pts

Express u' in terms of r , $\dot{r}$ and $\dot{\varphi}$. Find $u'(0)$ corresponding to the value of $\varphi = 0$.

Explicitly differentiate u :

$$u' = \frac{du}{d\varphi} = -\frac{1}{r^2} \frac{dr}{d\varphi}$$

Let us represent $\frac{dr}{d\varphi}$ как производную сложной функции:

$$\frac{dr}{d\varphi} = \frac{\dot{r}}{\dot{\varphi}}$$

from where:

The value is constant and equal to:

$$r^2\dot{\varphi} = \frac{L}{m} = v_0 b$$

At the initial moment, the speed is directed directly towards the wire. Then we find:

Answer: \textbf{Answer}:

$$u' = -\frac{\dot{r}}{r^2\dot{\varphi}}$$

C2

0.30 pts

Express the mechanical energy E of the ball in terms of m , u , u' , v_0 , b and A .

The expression for the mechanical energy of the ball is as follows:

$$E = \frac{m\dot{r}^2}{2} + \frac{mr^2\dot{\varphi}^2}{2} - \frac{A}{r^2}$$

From angular momentum conservation we have:

$$L = mr^2\dot{\varphi} = mv_0b$$

hence:

$$E = \frac{m\dot{r}^2}{2} + \frac{mv_0^2b^2}{2r^2} - \frac{A}{r^2}$$

Переходя к переменным u and u' , we obtain:

Answer: \textbf{Answer:}

$$\frac{mv_0^2b^2}{2}(u'^2 + u^2) - Au^2 = E = \frac{mv_0^2}{2}$$

C3

0.70 pts

Get addicted $u(\varphi)$. Express your answer in terms of m , v_0 , b , A and φ .

Differentiating the expression for energy with respect to φ , we obtain the equation of harmonic vibrations:

$$u'' = -u \left(1 - \frac{2A}{mv_0^2b^2} \right)$$

с циклической частотой:

$$\omega_0 = \sqrt{1 - \frac{2A}{mv_0^2b^2}}$$

Общее solution let us write в виде:

$$u(\varphi) = C \sin(\omega_0\varphi + \varphi_1)$$

Taking into account начальных условий:

$$u(0) = 0 \quad u'(0) = \frac{1}{b}$$

We find:

Answer: \textbf{Answer:}

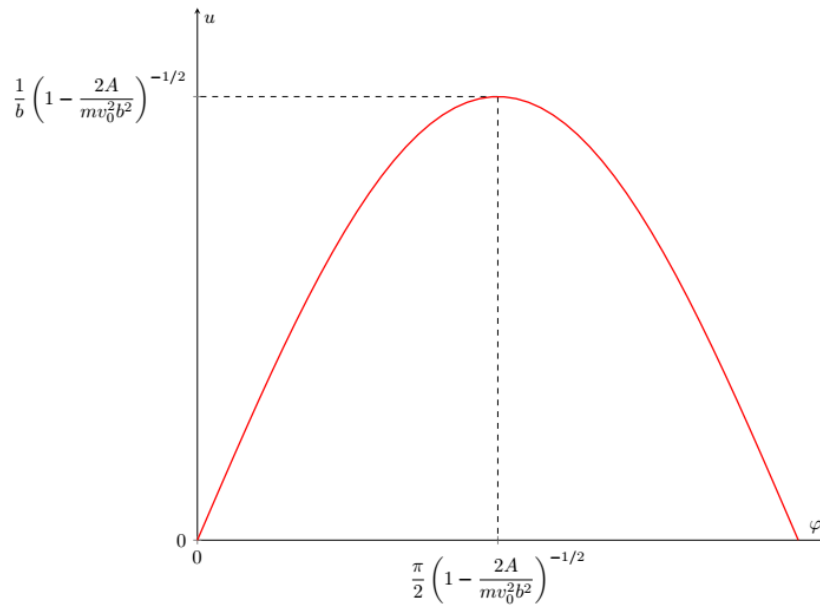
$$u(\varphi) = \frac{1}{b\sqrt{1 - \frac{2A}{mv_0^2b^2}}} \sin \left(\varphi \sqrt{1 - \frac{2A}{mv_0^2b^2}} \right)$$

C4

0.60 pts

On the answer sheet, qualitatively construct the trajectory of the center of the ball in coordinates u, φ . The coordinates of all feature points can be expressed in terms of m , v_0 , b and A .

The value u , starting from zero, reaches a maximum, and then, reaching zero, remains constant in time. The graph is shown in the figure below:



C5

0.50 pts

Using the value of the parameter A found in step **A4**, find the initial speed of the ball v_1 at which during the interaction between it and the wire the direction of the ball's speed changes to the opposite. Express your answer in terms of m , b , R , λ and the electrical constant ϵ_0 .

When the direction of the ball's speed is reversed, the total change in angle φ is 2π . From here we find the value ω_0 :

$$\omega_0 = \frac{1}{2}$$

Then parameter A equals:

$$A = \frac{3mv_1^2b^2}{8}$$

For/Аравнивая с expressionм, погауенным в parte **A4**

$$\frac{\lambda^2 R^3}{2\pi\epsilon_0} = \frac{3mv_1^2b^2}{8}$$

we find:

Answer: \textbf{Answer:}

$$v_1 = \frac{2\lambda R}{b} \sqrt{\frac{R}{3\pi\epsilon_0 m}}$$

D1

0.60 pts

Find S and q_2 . Express your answers using R , L and q .

Let us introduce a coordinate system at the center of a cylindrical sphere with the axis z directed along the line connecting the center of the sphere with the charge q_1 . The potential of charges q_1 and $-q_2$ at any point on the surface of the ball is equal to zero, therefore:

$$\frac{q_1}{\sqrt{r^2 + (x - L)^2}} = \frac{q_2}{\sqrt{r^2 + (x - S)^2}}$$

or:

$$(q_1^2 - q_2^2)(r^2 + x^2) + 2x(q_2^2 L - q_1^2 S) + q_1^2 S^2 - q_2^2 L^2 = 0$$

Since $r^2 + x^2 = R^2$, coefficient for/at x equals нулю. Then we get систему уравнений:

$$\frac{L}{S} = \frac{q_1^2}{q_2^2} \quad (q_1^2 - q_2^2)R^2 = q_2^2 L^2 - q_1^2 S^2$$

solving which, we find:

Answer: \textbf{Answer:}

$$S = \frac{R^2}{L} \quad q_2 = \frac{q_1 R}{L}$$

D2

0.40 pts

Find dq_1 , dq_2 , S and r_- . Express your answers using λ , R , r , α and $d\alpha$.

For dq_1 we have:

$$dq_1 = \lambda r d \tan \alpha$$

or:

The element in question is located at a distance $L = \frac{r}{\cos \alpha}$ from the center of the ball.

From here we find dq_2 and S :

For r_- we have:

$$r_- = r - S \cos \alpha$$

or

Answer: \textbf{Answer:}

$$dq_1 = \frac{\lambda r}{\cos^2 \alpha} d\alpha$$

D3

0.20 pts

Find the magnitude and direction (attraction or repulsion) of the vector sum of forces $d\vec{F} = d\vec{F}_+ + d\vec{F}_-$ interaction of charges dq_2 and $-dq_2$ with the wire. Express your answer in terms of λ , R , r , α , $d\alpha$ and the electrical constant ϵ_0 .

The pair of image charges are attracted to the wire because the negative charge is at a smaller distance from the wire. For the force of attraction, taking into account the electric field of the wire, we have:

$$dF = \frac{\lambda dq_2}{2\pi\epsilon_0} \left(\frac{1}{r_-} - \frac{1}{r} \right)$$

or:

Answer: \textbf{Answer:} The force of attraction is

$$dF = \frac{\lambda^2 R^3}{2\pi\epsilon_0 r^3} \frac{\cos \alpha d\alpha}{1 - \frac{R^2}{r^2} + \frac{R^2}{r^2} \sin^2 \alpha}$$

D4

1.00 pts

Find the magnitude and direction of the total force of interaction between the ball and the wire F_{noLH} . Express your answer in terms of λ , R , θ and the electric constant ϵ_0 .

To find the total force, we calculate the integral of the expression obtained in point **D3**

in the range from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$. Let's introduce the value t , equal to:

$$t = \frac{R \sin \alpha}{\sqrt{r^2 - R^2}}$$

Then we get:

$$F = \frac{\lambda^2 R^2}{2\pi\epsilon_0 r \sqrt{r^2 - R^2}} \int_{-\frac{R}{\sqrt{r^2 - R^2}}}^{\frac{R}{\sqrt{r^2 - R^2}}} \frac{dt}{1 + t^2} = \frac{\lambda^2 R}{\pi\epsilon_0 r} \frac{R}{\sqrt{r^2 - R^2}} \arctan \frac{R}{\sqrt{r^2 - R^2}}$$

Taking into account relation r , R and θ , we find:

Answer: \textbf{Answer:} The force of attraction is

$$F = \frac{\lambda^2 \sin^2 \theta \cdot \theta}{\pi\epsilon_0 \cos \theta}$$

D5

1.00 pts

Find the potential energy W_p of the interaction of the ball with the wire. Express your answer in terms of λ , R , θ and the electric constant ϵ_0 .

Considering that the ball is attracted to the wire, the expression for potential energy is:

$$W_p = - \int_r^\infty F(r) dr$$

Let us find quantity dr в переменных R and θ :

$$dr = d\left(\frac{R}{\sin \theta}\right) = -\frac{R \cos \theta d\theta}{\sin^2 \theta}$$

Then we have:

$$W_p = -\frac{\lambda^2 R}{\pi\epsilon_0} \int_0^\theta \theta d\theta$$

from where:

Answer: \textbf{Answer:}

$$W_p(\theta) = -\frac{\lambda^2 R \theta^2}{2\pi\epsilon_0}$$

D6

0.30 pts

What speed v_{sm} will an uncharged metal ball of radius R and mass m have when colliding with a wire if it is released from infinity without an initial speed? Express your answer in terms of λ , R , m and the electric constant ϵ_0 .

At the moment of contact between the ball and the wire $\theta = \frac{\pi}{2}$. Let's write down the law of conservation of energy:

$$\frac{mv_0^2}{2} = \frac{\lambda^2 R}{2\pi\epsilon_0} \left(\frac{\pi}{2}\right)^2$$

from where:

Answer: \textbf{Answer:}

$$v = \frac{\pi\lambda}{2} \sqrt{\frac{R}{\pi\epsilon_0 m}}$$

E1

0.40 pts

For a given value v_0 , obtain the dependence of the square of the radial velocity component $\dot{r}^2$ on the parameter θ . Express your answer in terms of m , v_0 , λ , R , b , electric constant ϵ_0 and θ .

From the law of conservation of energy we have:

$$\frac{mv^2}{2} - \frac{\lambda^2 R \theta^2}{2\pi\epsilon_0} = \frac{mv_0^2}{2}$$

Let us represent velocity в следующем виде:

$$v^2 = \dot{r}^2 + r^2 \dot{\phi}^2$$

Taking into account закона сохранения момента momentuma

$$L = mv_0 b = mr^2 \dot{\phi}$$

we find:

Answer: \textbf{Answer:}

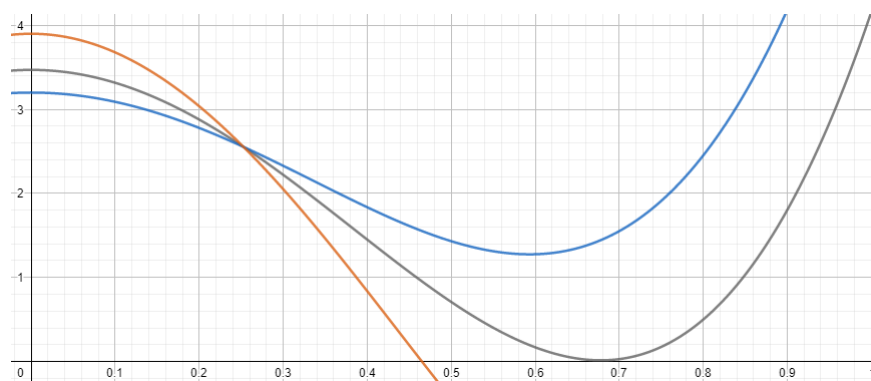
$$\dot{r}^2 = v_0^2 - v_0^2 \frac{b^2}{R^2} \sin^2 \theta + \frac{\lambda^2 R}{\pi\epsilon_0 m} \theta^2$$

E2

0.60 pts

Construct high-quality graphs of the dependence $\dot{r}^2(\theta)$ for different values of v_0 .

Depending on the value of v_0 , functions are divided into three types: 1) intersecting the ordinate axis at $\theta < \frac{\pi}{2}$; 2) ordinates that do not intersect the axis; 3) touching the ordinate axis at $\theta < \frac{\pi}{2}$.



E3

0.60 pts

Considering all quantities except v_0 constant, you obtain a system of equations that allows you to determine the value v_{crit} . The system of equations can include m , R , b , λ , electric constant ϵ_0 , θ_{max} and v_{crit} .

At sufficiently high speeds (type 1 function), the graph intersects the ordinate axis. At the moment of intersection, the value of $\dot{r}$ is reversed because $\dot{r}^2 > 0$. At sufficiently low

speeds (type 2 function), the value of $\dot{r}^2$ is positive at the extremum, so the ball falls on the wire. The function of the 3rd type is a boundary case of those considered above - at the extremum, the value of the function $\dot{r}^2$ becomes zero. At any speed, the ball will not touch the wire. From here it is clear that when the value $\dot{r}^2$ is equal to zero, the derivative $\frac{d\dot{r}^2}{d\varphi}$ must also be equal to zero, which is written in the form of a system as follows:

Answer: \textbf{Answer:}

$$\left(\frac{b^2 \sin^2 \theta_{max}}{R^2} - 1 \right) = \frac{\lambda^2 R \theta_{max}^2}{m v_0^2 \pi \epsilon_0} \quad \frac{b^2 \sin 2\theta_{max}}{R^2} = \frac{2\lambda^2 R \theta_{max}}{m v_0^2 \pi \epsilon_0}$$

E4

1.20 pts

Find v_{crit1} , v_{crit2} and v_{crit3} corresponding to the following values of b :

$$b_1 = 20R \quad b_2 = 3R \quad b_3 = 1,1R$$

Answers вычитаются via/through m, R, λ and электрическую постоянную ϵ_0 .

From the equation for the zero derivative we find:

$$v_0 = \lambda \sqrt{\frac{R}{\pi \epsilon_0 m}} \cdot \frac{R}{b} \sqrt{\frac{2\theta_{max}}{\sin 2\theta_{max}}}$$

For нахождения v_0 необходимо ограничить θ_{max} . Разделив уравнения друг на друга, we get:

$$\frac{k^2 \sin^2 \theta_{max} - 1}{k^2 \sin 2\theta_{max}} = \frac{\theta_{max}}{2} \quad k = \frac{b}{R}$$

Solving the equation на калькуляторе, we find:

$$\theta_{max1} = 0,296 \text{ rad} \quad \theta_{max2} = 0,793 \text{ rad} \quad \theta_{max3} = 1,459 \text{ rad}$$

from where:

Answer: \textbf{Answer:}

$$v_{crit1} \approx 5,15 \cdot 10^{-2} \lambda \sqrt{\frac{R}{\pi \epsilon_0 m}} \quad v_{crit2} \approx 3,15 \cdot 10^{-1} \lambda \sqrt{\frac{R}{\pi \epsilon_0 m}} \quad v_{crit3} \approx 3,30 \lambda \sqrt{\frac{R}{\pi \epsilon_0 m}}$$